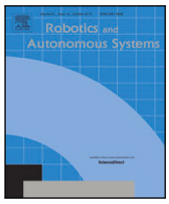
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Model predictive variable impedance control towards safe robotic interaction in unknown disturbance-rich environments

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ABSTRACT

Robotic systems have evolved to handle various significant interaction tasks in different environments. Under these conditions, the involvement of humans in the environment drastically complicates such interaction tasks; as the safety of humans should be prioritized while seeking to achieve the desired task aim. It is thus paramount that appropriate developments should be pursued with specific considerations for such safety-performance-balanced interaction tasks on unknown soft environments (e.g., humans). Towards this end, we present an adaptive robust, and passive control scheme based on model predictive control and variable impedance control that addresses this challenge. Under this control scheme, during robotic interaction tasks with complex environments (e.g., humans), the presented development and design incorporate safety thresholds that are carefully satisfied via impedance adaptation, and realized by a safety-related mode-switching mechanism. Once the safety thresholds are satisfied, task performance is then focused on. Additionally, a real-time adaptive robust parameter estimator is designed and utilized to estimate the environment contact model for the model predictive control, and thus this control scheme is robust against disturbances (e.g., which would invariably arise from the inevitable small bounded human motions) during the interaction tasks. Finally, the key safety and performance attainments of the proposed control scheme are verified via experiments. The experiments are conducted on two silicone rubber models and a human arm. These show that the proposed control scheme effectively outperformed various currently available control schemes in these interaction tasks with unknown environment contact models, and bounded but unpredictable environment position shifts, such as in robotic ultrasound scanning applications.

1. Introduction

After the introduction of industrial robots in the past century, the field of robotics has been trending towards solving interaction tasks in unstructured environments. The rise of human–robot collaboration and interaction have further propelled this direction through means such as collaborating in factories to learn and optimize the assembly task [1], assisting in medical/surgical operation [2], and rehabilitation [3], etc. However, developing control laws with a safe interaction between humans and robots is nontrivial.

Impedance control (IC) has been widely adopted since its introduction in [4]. As an indirect force control method, IC regulates the robot–environment interaction based on a virtual impedance model, enabling robots to comply with varying contact forces and avoiding risks of damaging the environment during interaction tasks compared to direct force control methods. As a growing number of tasks require a flexible impedance model, variable impedance control (VIC) is thus designed, which changes the impedance model in the control loop for better adaptivity [5]. Many previous works achieve VIC via learning-based methods such as learning from demonstration [6–9] and reinforcement learning [10–12], but they are respectively limited by the arduous demonstration process or the sim-to-real problem. In [13,14], linear quadratic regulators (LQRs) generate the optimal impedance parameters via human demonstrations. In [15], the impedance parameters are tuned based on dynamics models, and sliding mode control is adopted for handling interaction uncertainties. The impedance variation can also be realized via bio-mimetic methods for motor behavior learning [16], dual-arm operation [17], and manipulator control for contact tasks [18]. Bio-mimetic methods humanize the robotic interaction and enhance user experiences, but they might increase the computational complexity and reduce the generalizability.

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However, neither IC nor VIC inherently incorporates any constraints, potentially leading to failures in tasks requiring the robotic system to obey constraints [19,20]. Model predictive control (MPC), initially developed for the chemical process industry in the late 1970s [21], has become a widely adopted technique [22] due to its ability to optimize subject to multiple constraints. In recent years, MPC has also been favored for interaction control, thanks to its optimization with constraints [23]. Therefore, MPC can be considered for integration with IC or VIC to complement their functionalities and address the constraint issues.

Several works have successfully integrated MPC with IC or VIC, demonstrating the effectiveness of this combined approach in robotic interaction tasks. In [24], an admittance control is designed based on MPC with constraints of velocity and interaction force. In [25], the objective function for MPC is constructed based on the desired impedance model with user-designed constraints. Furthermore, the MPC is adopted in [26] to optimize the impedance parameters for tracking error minimization. In [27], a learned model for MPC adapts the impedance parameters to realize the VIC. In [28], the MPC is employed to optimize the robot trajectory and impedance model online via a task model from demonstrations. All the aforementioned works design the objective concisely, e.g., minimizing tracking errors, but they may not satisfy the complicated task requirements. For instance, in a human-involved interaction task, the control objective should prioritize humans' safety and then achieve the desired control. No previous works have proposed such complex objectives for optimization with constraints.

To address this research gap, a novel model predictive variable impedance control (MPVIC) is proposed in this work. The proposed MPVIC aims at maintaining safety and then achieving a desired performance, i.e., safety-performance balance, with constraint compliance, robustness, and passivity guarantee, for an interaction task with unknown soft environments, e.g., humans. One real-world example is diagnostic ultrasonography, where a doctor holds a probe and moves it along the human skin to obtain an ultrasonic image inside the body, as illustrated in Fig. 1.

In the proposed MPVIC, passivity-based control is first employed to ensure safety and stability during interactions between robots and humans as well as to adapt to the changes in system dynamics. Passivity describes how a system handles energy: a system is passive if its inflow of energy is always non-negative, i.e., it never creates energy on its own. In [29], a passive variable impedance control is proposed based on electromyography for operator intention detection. In [30], a passive impedance control scheme is proposed to control the maximum penetration depth during the robotic ultrasound scanning task. In [31], the energy-aware variable impedance control is proposed for aerial vehicles in the port Hamiltonian framework to achieve free-flight and in-contact stability. To guarantee the passivity of MPVIC, the tank-based approach inspired by [32], is designed in this work. In this approach, energy dissipated from the system is accumulated in a virtual energy tank, ensuring the overall system remains passive.

Furthermore, these interaction tasks present significant challenges due to the inherent properties of soft environments. The soft environments are generally nonlinear, viscoelastic, and inhomogeneous, further compounded by the high degree of uncertainty and disturbance due to the unpredictable nature of human movements. It is also noteworthy that the performance of MPC relies on the accuracy of the environment model. For contacts with a soft environment, three commonly used linear viscoelastic models are the Maxwell model, Kelvin–Voigt model, and Zener model [33]. Nevertheless, the Hunt–Crossley (HC) model introduced in [34] offers a better fit for soft environments and human tissues such as membranes, muscles, and skins [35,36]. To address the challenges of interacting with soft environments and ensuring accurate model in MPC, a novel adaptive robust parameter estimator with a dead zone is integrated into this work, such that the estimator, originally for linear models in [37], is then able to estimate the nonlinear HC model accurately in real-time during the interaction between the robot and environment.

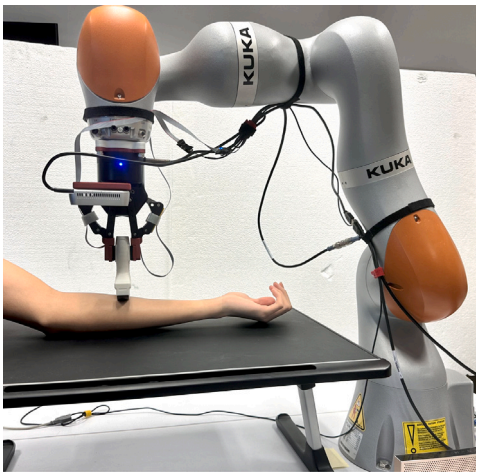


Fig. 1. A robotic interaction task with a human.

The main contributions of this work are:

- development of a novel MPVIC scheme with specific considerations for safety-performance-balanced interaction tasks on unknown soft environments, e.g., humans;
- an adaptive robust parameter estimator for real-time nonlinear environment model estimation;
- a theoretical analysis that guarantees the passivity of the control scheme (ensuring safety) as well as the convergence and robustness of the environment estimator (demonstrating system adaptability);
- appropriate experiments (conducted on two silicone rubber models and a human arm) which show that the proposed control scheme effectively outperformed various currently available control schemes in these interaction tasks with unknown environment contact models, and bounded but unpredictable environment position shifts, such as in robotic ultrasound scanning applications.

The rest of this paper is organized as follows. Section 2 formulates the problem to be solved. Following that, the proposed control scheme is presented in detail in Section 3. Section 4 shows and discusses the experimental results of the proposed control scheme along with a comparison to two state-of-the-art control schemes. Finally, Section 5 provides the conclusion.

2. Problem formulation

This work focuses on one category of physical human–robot interaction tasks, examples of which are ultrasonography, body massage, etc. Specifically, this category of tasks aims at safely applying a suitable contact force on humans while moving along its surface. As mentioned in Section 1, such human environments are extremely complex: they are unknown, nonlinear, viscoelastic, inhomogeneous tangentially and normally, and with unpredictable disturbance. In this work, the safety of interaction is first emphasized by the control scheme, and then the desired impedance behavior is maintained. To further enhance the safety of interaction, constraints on position, velocity, and acceleration can be enabled via MPC. For a successful MPC with unknown or inhomogeneous environments, an environment model that accounts for disturbances is crucial. To address these challenges, the following definitions and assumptions are established.

Definition 1. The interaction is defined to be safe when the force and its first-order time derivative are both within certain predefined ranges and vice versa.

Assumption 1. Interaction with the environment is within the robot workspace without singularity. The initial position and orientation of the environment are known to the robot, which starts in contact with the environment.

Assumption 2. The disturbance $\delta(t)$, e.g., environment position shift, sensor noise, etc., is bounded by Δ as in (1), and the environment orientation is invariant in the task.

$$\sup|\delta(t)| \leq \Delta, \tag{1}$$

Definition 1 narrows down the definition of safety by ISO/TS 15066 [38] into the force limit only for simplicity without the loss of generality, as the velocity and power of the robot are insignificant in a slow interaction task and can always be added into the safety-related mode design presented later. Assumption 1 can be released via path planning and visual recognition, which are beyond the scope of this work. Assumption 2 is realistic as the human is expected to stay near the initial position during the task, and large motions, if any, can be estimated as part of the environment, or compensated by additional visual feedback, which is beyond the scope of this work as well.

This work focuses on the impedance control along the z direction, i.e., the normal direction of the environment without the loss of generality, because the same control method can be applied to any direction. Initially, consider the dynamics of a robot manipulator:

$$M(q)\ddot{q} + C(q,\dot{q}) + G(q) = \tau_c - J^T(q) f_{env}, \tag{2}$$

where q is the joint variable of the manipulator, τ_c is the control torque of the manipulator, $M(q)$, $C(q,\dot{q})$, $G(q)$ are the known dynamics matrices, J is the Jacobian matrix, and f_{env} is the interaction force. Define a matrix D:

$$D = \begin{bmatrix} & 0_{2\times6} & \\ r_{31} \quad r_{32} & r_{33} & 0 \quad 0 \quad 0 \\ & 0_{3\times6} & \end{bmatrix}, \tag{3}$$

where r_{31}, r_{32}, and r_{33} are elements from the third row of the 3×3 rotational matrix R between the robot base frame and the task frame, and $0_{r\times c}$ denotes an r-by-c zero matrix. With Cartesian variable p and the relationship $\ddot{p} = J\ddot{q} + \dot{J}\dot{q}$, one obtains:

$$\ddot{z} = D\ddot{p} = D(J\ddot{q} + \dot{J}\dot{q}). \tag{4}$$

$$\ddot{q} = J^{-1}D^{+}(\ddot{z} - D\dot{J}\dot{q}), \tag{5}$$

where D^+ denotes the pseudo-inverse of matrix D. The low-level control law along the z direction, namely τ_{cz}, is obtained by taking (5) back to (2):

$$\tau_{cz} = M(q)J^{-1}D^{+}(\ddot{z} - D\dot{J}\dot{q}) + C(q,\dot{q}) + G(q) + J^T f_{env}, \tag{6}$$

With the reference position z_r and the reference force f_r, the position error e_z and the force error e_f are defined as:

$$e_z = z_r - z, \quad e_f = f_r - f_{env}. \tag{7}$$

The interaction task in this paper aims to maintain stable contact with the environment, i.e., $\dot{z}_r = \ddot{z}_r = 0$. The coordinates of robot end-effector position z and environment force f_{env} are defined in Fig. 2.

The impedance model for VIC is then formulated as:

$$m_i\ddot{e}_z + d_i(t)\dot{e}_z + k_i(t)e_z = e_f, \tag{8}$$

where m_i, $d_i(t)$, and $k_i(t)$ are the impedance parameters for VIC, among which only the damping and stiffness coefficient are adapted, and the mass coefficient remains constant. The use of force error e_f in the impedance model ensures that the robot motion adapts to achieve the desired force profile, which is particularly suitable for tasks requiring accurate force control and compliance as discussed in this paper.

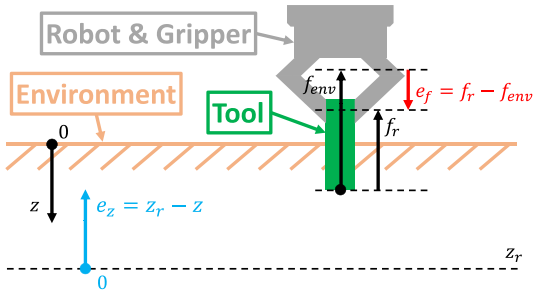


Fig. 2. Coordinate definitions for position and force.

Different from the majority of previous MPC-based VIC works such as [26], the environment interaction force model is considered in the MPC model here for a better prediction accuracy of MPC. Consider thus both the contact situation ($\hat{f}_{HC} > 0$) and the non-contact situation ($\hat{f}_{HC} = 0$); the environment force estimated from the HC model $\hat{f}_{HC}(z,\dot{z})$ is:

$$\hat{f}_{HC}(z,\dot{z}) = \max(kz^n + cz^n\dot{z},\ 0), \tag{9}$$

where function $\max(a,b)$ outputs the larger value from a or b, and k, c, and n are the HC model parameters, which are estimated in real-time during the task by the parameter estimator. Rewrite (9) with (7), and it results in:

$$\hat{f}_{HC}(e_z,\dot{e}_z) = \max((k - c\dot{e}_z)(z_r - e_z)^n,\ 0), \tag{10}$$

Focusing on the contact case where $\hat{f}_{HC} > 0$, the nonlinear system equation is obtained by substituting (10) into (8):

$$m_i\ddot{e}_z + [d_i(t) - c(z_r - e_z)^n]\dot{e}_z + k_i(t)e_z + k(z_r - e_z)^n - f_r = 0. \tag{11}$$

Regarding the time-varying damping and stiffness coefficients for VIC, when the interaction is considered unsafe, the damping should increase while the stiffness should decrease. In this work, damping and stiffness parameters d_i, k_i are adapted, and the desired damping and stiffness are $\{d_{i,min}, k_{i,max}\}$, while $\{d_{i,max}, k_{i,min} = 0\}$ are adopted under unsafe conditions. A smooth transition of VIC coefficients and the passivity guarantee for the task are also necessary.

3. Control scheme

To solve the control problem formulated in the previous section, a control scheme is proposed here that incorporates (*i*) the real-time adaptive robust parameter estimator to estimate the HC contact force model, and (*ii*) the MPVIC scheme that controls and regulates the robot following the desired impedance behavior under safe scenarios. The general structure of the proposed control scheme is shown in Fig. 3, with details as in the following subsections.

3.1. Adaptive robust parameter estimator

An adaptive robust parameter estimator is designed to estimate the HC contact force model $\hat{f}_{HC}(t)$ based on actual force feedback $f_{env}(t)$. The nonlinear HC model is initially linearized via logarithmic operation, and the parameter estimator output $\hat{y}(t) = \log(\hat{f}_{HC}(t))$ tracks the optimal output $y(t) = \log(f_{env}(t))$, which can be rewritten in the

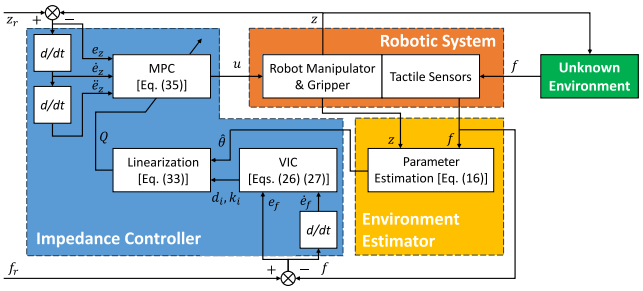


Fig. 3. Block diagram of the proposed MPVIC scheme.

linear-in-the-parameter form as:

$$\begin{aligned}\hat{y}(t) &= \log([k(t-1)+c(t-1)\dot{z}(t-1)]z(t-1)^{n(t-1)})\\ &= \log(k(t-1)) + n(t-1)\log(z(t-1))\\ &\quad + \log(1+\frac{c(t-1)\dot{z}(t-1)}{k(t-1)})\\ &= \log(k(t-1)) + n(t-1)\log(z(t-1))\\ &\quad + \sum_{i=1}^{\infty}\frac{(-1)^{i+1}(c(t-1)\dot{z}(t-1))^i}{ik(t-1)^i}\\ &\approx \hat{\theta}(t-1)^T\phi(t-1),\end{aligned} \tag{12}$$

where $\hat{\theta}(t)$ and $\phi(t)$ are defined as:

$$\begin{aligned}\hat{\theta}(t) &= [\log(k(t)), n(t), \frac{c(t)}{k(t)}]^T,\\ \phi(t) &= [1, \log(z(t)), \dot{z}(t)]^T.\end{aligned} \tag{13}$$

Here, $\hat{\theta}(t)$ is the vector that includes all parameters to be estimated, and $\phi(t)$ is the vector that includes remaining terms for $\hat{\theta}(t)$. The approximation in (12) holds based on the assumption as follows.

Assumption 3. The viscosity term $c(t-1)\dot{z}(t-1)$ in the HC model is much smaller than the stiffness term $k(t-1)$.

This assumption holds for most real-world environments; a situation which is also evidenced in the experiments here, and with real-world conditions which have been experimentally verified. For those extremely viscous environments, a different contact force model can be considered; though naturally with this much more challenging condition (of extremely viscous environments), there will be associated costs of possibly reduced prediction accuracy and increased estimation frequencies. Based on Assumption 3, only the first-order term remains in the summation of (12). The estimation error is defined as:

$$e_y(t) = y(t) - \hat{y}(t). \tag{14}$$

To overcome the disturbance δ bounded by Δ in (1), a dead zone is defined as:

$$a(t-1) = \begin{cases} 1, & |e_y(t)| > 2\Delta \\ 0, & |e_y(t)| \le 2\Delta. \end{cases} \tag{15}$$

It is noteworthy that the disturbance upper bound Δ should be carefully selected to balance between the estimation accuracy and the computational consumption. The parameter estimator is then designed with (14) and (15):

$$\hat{\theta}(t) = \hat{\theta}(t-1) + \frac{a(t-1)\phi(t-1)[y(t) - \hat{\theta}(t-1)^T\phi(t-1)]}{1+|\phi(t-1)|^2}. \tag{16}$$

For the developments above, the following intermediate result (by way of a stated Lemma) can be established and noted at this point.

Lemma 1. *With the parameter estimator (16), all estimated parameters in $\hat{\theta}(t)$ converge to their true values with robustness against a disturbance δ bounded by Δ.*

Proof of Lemma 1. Define the true parameter vector as θ^*. The actual force feedback can then be expressed with θ^* as:

$$y(t) = \theta^{*T}\phi(t-1) + \delta(t). \tag{17}$$

Define the parameter estimation error $\tilde{\theta}(t)$ as:

$$\tilde{\theta}(t) = \theta^* - \hat{\theta}(t). \tag{18}$$

Substitute (17) and (18) into (16), one obtains:

$$\tilde{\theta}(t) = \tilde{\theta}(t-1) - \frac{a(t-1)\phi(t-1)e_y(t)}{1+|\phi(t-1)|^2}. \tag{19}$$

Square both sides of (19) results in:

$$\begin{aligned}|\tilde{\theta}(t)|^2 &= |\tilde{\theta}(t-1)|^2 - \frac{2a(t-1)[e_y(t)-\delta(t)]e_y(t)}{1+|\phi(t-1)|^2}\\ &\quad + \frac{a(t-1)^2|\phi(t-1)|^2e_y(t)^2}{[1+|\phi(t-1)|^2]^2}\\ &\le |\tilde{\theta}(t-1)|^2 - \frac{2a(t-1)[e_y(t)-\delta(t)]e_y(t)}{1+|\phi(t-1)|^2}\\ &\quad + \frac{a(t-1)e_y(t)^2}{1+|\phi(t-1)|^2}\\ &= |\tilde{\theta}(t-1)|^2 - \frac{a(t-1)e_y(t)[e_y(t)-2\delta(t)]}{1+|\phi(t-1)|^2}.\end{aligned} \tag{20}$$

From the definition of the dead zone in (15), one obtains:

$$a(t-1) = \begin{cases} 0, & e_y(t)[e_y(t)-2\delta(t)] \le 0 \\ 1, & e_y(t)[e_y(t)-2\delta(t)] > 0, \end{cases} \tag{21}$$

which results in:

$$a(t-1)e_y(t)[e_y(t)-2\delta(t)] \ge 0. \tag{22}$$

Take (22) back to (20), the following properties are obtained:

$$\begin{aligned}&|\tilde{\theta}(t)|^2 - |\tilde{\theta}(t-1)|^2 \le 0,\\ &|\theta^* - \hat{\theta}(t)| \le |\theta^* - \hat{\theta}(t-1)| \le |\theta^* - \hat{\theta}(0)|.\end{aligned} \tag{23}$$

The convergence of the estimator is thus proved in (23). □

Remark 1. The parameter estimator (16) is robust and will adapt when the estimation error $e_y(t)$ exceeds twice the disturbance bound Δ due to the dead zone (15).

Remark 2. The parameter estimator (16) is able to estimate the environment when the disturbance δ does not exceed the predefined upper bound Δ. Otherwise, the estimator regards δ as part of the signal $y(t)$. The temporarily wrong parameter estimations do not affect the control performance because they treat the bound-exceeding disturbance as part of an 'updated environment' and match $\hat{y}(t)$ with $y(t)$, and correct parameter estimations are retrieved when δ reduces back within Δ.

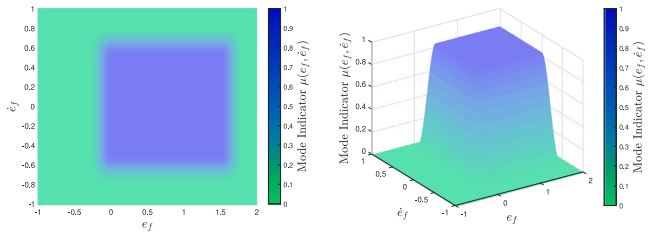
3.2. Model predictive variable impedance control

Firstly, two modes for the proposed MPVIC are defined: safety mode and performance mode, with user-defined impedance model parameters $\{d_{i,min}, k_{i,max}\}$ (desired impedance model for performance mode that achieves the performance) and $\{d_{i,max}, k_{i,min}\}$ (conservative impedance model for safety mode that avoid dangers). In the safety mode, the safety of the environment, e.g., the human, is exceedingly emphasized by sacrificing the task performance, reducing the stiffness to $k_{i,min}$ while increasing the damping to $d_{i,max}$ for the impedance model. In the performance mode, the robot focuses on the task performance by following the desired impedance model and adopting impedance parameters $\{d_{i,min}, k_{i,max}\}$.

Based on Definition 1 in Section 2, the controlled system enters the safety mode only upon abnormal force feedback. Thus, two safety thresholds are defined in terms of interaction force error e_f and its

Table 1
Definition of modes and impedance parameters.

Mode	Thresholds of e_f, $\dot{e}_f$	$k_i(t)$	$d_i(t)$
Safety	$e_f \in (-\infty, -\omega_1] \cup [f_r, +\infty)$, $\dot{e}_f \in (-\infty, -\dot{e}_{fb}] \cup [\dot{e}_{fb}, +\infty)$	0	$d_{i,max}$
Performance	$e_f \in [0, f_r - \omega_1]$, $\dot{e}_f \in [-\dot{e}_{fb} + \omega_2, \dot{e}_{fb} - \omega_2]$	$k_{i,max}$	$d_{i,min}$
Transition	Otherwise	$(0, k_{i,max})$	$(d_{i,min}, d_{i,max})$



(a) (b)

Fig. 4. One example of the safety, transition, and performance modes: (a) top view; (b) 3D view.

time derivative $\dot{e}_f$. Safety mode is entered at the beginning of the task as well as when safety thresholds are violated. Performance mode is entered when safety thresholds are satisfied. To ensure a smooth transition between the two modes, a third mode named transition mode is designed, which links the two modes with smooth impedance variations. The boundaries and values of different modes are defined in Table 1, where ω_1 and ω_2 are transition mode widths.

Smooth transitions between the safety mode and transition mode, and between the transition mode and performance mode are essential for both the safety and the performance of the task [39]. A transition function $\sigma(x, a, b)$ with transition width $\omega = b - a$ is then designed:

$$\sigma(x,a,b) = \begin{cases} 0, & x < a \\ \frac{1}{2} + \frac{1}{2}\sin(\frac{\pi}{\omega}(x-b) + \frac{\pi}{2}), & x \in [a,b] \\ 1, & x > b. \end{cases} \tag{24}$$

This transition function can transit between two modes smoothly. Based on (24) and Table 1, the mode indicator $\mu(e_f, \dot{e}_f)$ is constructed as:

$$\begin{aligned} \mu(e_f, \dot{e}_f) =& \sigma(e_f, -\omega_1, 0)[1 - \sigma(e_f, f_r - \omega_1, f_r)] \\ & \sigma(\dot{e}_f, -\dot{e}_{fb}, -\dot{e}_{fb} + \omega_2)[1 - \sigma(\dot{e}_f, \dot{e}_{fb} - \omega_2, \dot{e}_{fb})], \end{aligned} \tag{25}$$

where ω_1 and ω_2 are widths of the transition zone along e_f and $\dot{e}_f$, and $\dot{e}_{fb}$ indicates the boundaries. It is pertinent to note that selecting these width and boundary values is a matter that should be thoughtfully addressed. The normal approach is to refer to those from an exemplary human operation to decide the boundaries, and then select a sufficiently large transition width to ensure a smooth mode transition. The mode indicator $\mu(e_f, \dot{e}_f)$ indicates the safety mode when $\mu = 0$, the transition mode when $0 < \mu < 1$, and the performance mode when $\mu = 1$. An example of the distribution of the three modes is illustrated in Fig. 4.

With the above mode indicator, variable impedance controller parameters $k_i(t)$ and $d_i(t)$ are then designed with (25) as:

$$\begin{aligned} k_i(t) =& \mu(e_f, \dot{e}_f) k_{i,max}, \\ d_i(t) =& d_{i,max} + \mu(e_f, \dot{e}_f)(d_{i,min} - d_{i,max}). \end{aligned} \tag{26}$$

Because the increase of controller parameter $k_i(t)$ in (26) will result in the self-generation of energy in the controlled system, $k_i(t)$ is further restricted by the constrained stiffness. Define the constrained stiffness as $\bar{k}_i(t)$. Here, $\bar{k}_i(t)$ is initialized as the same value as unconstrained stiffness $k_i(0)$, and it is bounded in the non-negative real value for all time:

$$\bar{k}_i(0) = k_i(0), \quad \bar{k}_i(t) \geq 0 \quad \forall t. \tag{27}$$

Stiffness constraining is achieved via constraining the positive changing rate of $\bar{k}_i(t)$ to ensure the passivity:

$$\dot{\bar{k}}_i(t) = \begin{cases} \dot{k}_i(t) + \lambda e_{ki}(t), & |e_z| \leq \epsilon \\ \min(\dot{k}_i(t) + \lambda e_{ki}(t), \frac{2[d_i(t)\dot{e}_z^2 - f_r \dot{e}_z + p_1]}{e_z^2}), & |e_z| > \epsilon \end{cases} \tag{28}$$

where $e_{ki}(t) = k_i(t) - \bar{k}_i(t)$ is the difference between the theoretical unconstrained stiffness $k_i(t)$ and the actual constrained stiffness $\bar{k}_i(t)$, λ is a positive constant gain for tracking $\bar{k}_i(t)$ to $k_i(t)$. ϵ is a small positive threshold constant to avoid singularity when $e_z \to 0$. p_1 is defined as:

$$p_1 = \begin{cases} 0, & V_{tk} \leq 0 \\ \frac{\lambda e_{ki} e_z^2}{2}, & V_{tk} > 0. \end{cases} \tag{29}$$

p_1 here is the energy flow supported by a virtual energy tank V_{tk}. Essentially, V_{tk} provides the energy for the constrained stiffness $\bar{k}_i(t)$ to track the desired theoretical unconstrained stiffness $k_i(t)$. Here, V_{tk} is defined with a positive user-designed initial energy budget $V_{tk,0}$:

$$\begin{aligned} & V_{tk}|_{t=0} = V_{tk,0}, \\ & \dot{V}_{tk} = (d_i(t) - p_2)\dot{e}_z^2 - f_r \dot{e}_z - \frac{1}{2}\dot{\bar{k}}_i(t) e_z^2, \end{aligned} \tag{30}$$

where p_2 is a small positive constant to ensure the strict passivity when $\dot{e}_z \neq 0$.

Remark 3. The energy budget $V_{tk,0}$ should be appropriately selected based on the impedance model and environment model. Assuming that a nominal HC model is given for the interaction environment, the impedance model energy can be simulated during the interaction. The maximum energy can then be referred for deciding $V_{tk,0}$, such that it is neither so small that the controller becomes over-conservative, nor so large that the controller becomes too aggressive to effectively constrain the system behavior.

It will be rigorously proved later (with a result that we denote as Theorem 1) that p_1 is the upper bound of the energy flow into the human–robot system due to the constrained stiffness $\bar{k}_i(t)$ when $|e_z| > \epsilon$, and that V_{tk} ensures an upper bound of the energy in the system, i.e., the passivity. It will also be proved that when the energy budget in V_{tk} is fully consumed, i.e., $V_{tk} = 0$, the passive controller under (28) obeys passivity via reducing the stiffness $\bar{k}_i(t)$ and thus releasing energy back to V_{tk}.

Significantly, an extreme case under this design is that both energy stored in the energy tank and potential energy in the model stiffness are depleted, i.e., $V_{tk} = 0$, $\bar{k}_i(t) = 0$. A safety stop indicator s_{stop} is designed to stop the system execution under this extreme case:

$$s_{stop} = \begin{cases} \text{true} & V_{tk} < \epsilon, \text{ and } \bar{k}_i(t) < \epsilon, \\ \text{false} & \text{otherwise.} \end{cases} \tag{31}$$

Here, the controller will examine the value of s_{stop} in real-time and immediately stop the robot if the value is 'true' for safety considerations.

Based on $d_i(t)$ from (26) and $\bar{k}_i(t)$ from (28), the framework from [25] is improved by linearizing the nonlinear MPC for faster computation and thus faster response. Rewrite the system Eq. (11) in the function form:

$$\begin{aligned} f_{sys}(\ddot{e}_z, \dot{e}_z, e_z) =& m_i \ddot{e}_z + [d_i(t) - c(z_r - e_z)^n]\dot{e}_z \\ & + \bar{k}_i(t) e_z + k(z_r - e_z)^n - f_r. \end{aligned} \tag{32}$$

The linearization of (32) via first-order Taylor expansion is conducted at $e_z = e_{z0}$, $\dot{e}_z = \dot{e}_{z0}$, $\ddot{e}_z = \ddot{e}_{z0}$ when $t = t_0$:

$$f_{sys}(\ddot{e}_z, \dot{e}_z, e_z) \approx f_{sys}(\ddot{e}_{z0}, \dot{e}_{z0}, e_{z0}) + \left.\frac{\partial f_{sys}}{\partial \ddot{e}_z}\right|_{t_0} (\ddot{e}_z - \ddot{e}_{z0})$$

$$+\left.\frac{\partial f_{sys}}{\partial \dot{e}_z}\right|_{t_0}(\dot{e}_z-\dot{e}_{z0})+\left.\frac{\partial f_{sys}}{\partial e_z}\right|_{t_0}(e_z-e_{z0}). \tag{33}$$

Based on (33), the system equation is linearized as:

$$c_1\ddot{e}_z+c_2\dot{e}_z+c_3e_z+c_4=0, \tag{34}$$

where

$$\begin{aligned}c_1 &= m_i,\\ c_2 &= d_i(t_0)-c(z_r-e_{z0})^n,\\ c_3 &= \bar{k}_i(t_0)-n(k-c\dot{e}_{z0})(z_r-e_{z0})^{n-1},\\ c_4 &= -f_r+k(z_r-e_{z0})^n+n(k-c\dot{e}_{z0})(z_r-e_{z0})^{n-1}e_{z0}.\end{aligned}$$

With (34), define an objective function J with the discrete time $t_d=1,2,\ldots\in\mathbb{N}^+$ over a prediction horizon h_p with the MPC state x and the weight matrix Q that includes the impedance model information:

$$\begin{aligned}J &= \sum_{t_d}^{t_d+h_p} x^TQx,\\ x &= [\ddot{e}_z,\dot{e}_z,e_z,1]^T,\\ Q &= [c_1,c_2,c_3,c_4]^T[c_1,c_2,c_3,c_4].\end{aligned} \tag{35}$$

The linearized MPC problem is thus constructed in discrete time with the time interval T_s as:

$$\min_{u(t_d)} J=\min_{u(t_d)}\sum_{t_d}^{t_d+h_p}x^TQx$$
$$s.t.\begin{cases}x(t_d+1)=\begin{bmatrix}1&0&0&0\\T_s&1&0&0\\0&T_s&1&0\\0&0&0&1\end{bmatrix}x(t_d)+\begin{bmatrix}T_s\\0\\0\\0\end{bmatrix}u(t_d)\\ \text{other feasible constraints (See Remark 4.)}\end{cases} \tag{36}$$

Remark 4. The property of MPC, i.e., optimization along the objective function J subject to multiple constraints, provides freedom to design soft or hard MPC state or input constraints for different purposes such as safety consideration, as long as the MPC remains feasible. Specifically, as presented later in this work, we designed constraints in terms of position (a state constraint) to avoid damage to the environment.

Remark 5. The input signal $u(t_d)$ in (36) is the jerk of position error $\dddot{e}_z$. To realize the low-level control in (6), z̈ can be obtained as the time integral of $u(t_d)$.

Remark 6. The advantage of linearized MPC is confirmed by comparing the running time of nonlinear MPIC and linearized MPIC for 2000 time steps in MATLAB simulation on the same computer. The comparison study shows that the linearized MPC offers a significant computational advantage, requiring only 2.8 s compared to 23.2 s for nonlinear MPC. This translates to roughly eight times faster execution, which is crucial for meeting the real-time requirement of the system operating at 100 Hz. The model error introduced by the linearization process is negligible within the high-frequency framework, further justifying the use of linearized MPC. It is also noteworthy that a higher frequency, though demanding more stringent requirements in terms of communication and computation, would be beneficial for interaction scenarios (where applicable and needed) with faster interaction speed or higher requirements of interaction force sensitivity.

With all the developments above, the following main result can now be stated.

Theorem 1. *Given the human–robot system (2) performing a physical interaction task along the z-direction, controlled by the proposed MPVIC scheme (36) with environment model estimator (16) and variable impedance $d_i(t)$ in (26) and $\bar{k}_i(t)$ in (28), the following properties hold:*

1. *Safety-Performance Balance: The control system prioritizes the safety of the environment by regulating contact forces and achieves target impedance performance subject to arbitrary feasible constraints.*
2. *Passivity: The system remains passive with an energy upper bound for all time due to the tank-based passive control scheme, and the energy flow into the controlled system is regulated by p_1 in (29) when $|e_z|>\epsilon$.*

Proof of Theorem 1. At first, we prove the safe-performance balance of the proposed MPVIC. In the control scheme, MPC optimizes along the objective function J such that the system (34) holds with feasible MPC constraints. Based on (25), under unsafe scenarios in Table 1, the system switches to the safety mode $\mu(e_f,\dot{e}_f)=0$. According to (26) and (28), the conservative impedance model $\{m_i,d_{i,max},0\}$ is adopted under unsafe scenarios. On the contrary, the system switches to the performance mode $\mu(e_f,\dot{e}_f)=1$ when safety thresholds in Table 1 are satisfied, and the system then adopts the desired impedance model $\{m_i,d_{i,min},k_{i,max}\}$. This proves that MPVIC prioritizes the safety of the environment and then achieves target impedance performance.

Next, we prove the passivity of the human–robot system. Recall that a system is passive if its inflow of power is always non-negative. Consider the storage function that incorporates the kinetic energy and elastic energy of the impedance model:

$$V=\frac{1}{2}m_i\dot{e}_z^2+\frac{1}{2}\bar{k}_i(t)e_z^2. \tag{37}$$

Take the time derivative of (37), and one obtains:

$$\dot{V}=m_i\dot{e}_z\ddot{e}_z+\bar{k}_i(t)e_z\dot{e}_z+\frac{1}{2}\dot{\bar{k}}_i(t)e_z^2. \tag{38}$$

Substitute (11) into (38), which results in:

$$\begin{aligned}\dot{V}=&-[d_i(t)-c(z_r-e_z)^n]\dot{e}_z^2-k(z_r-e_z)^n\dot{e}_z\\&+f_r\dot{e}_z+\frac{1}{2}\dot{\bar{k}}_i(t)e_z^2.\end{aligned} \tag{39}$$

The energy that the robot injects into the system is denoted as V_{rob}, and the power is thus $\dot{V}_{rob}=f_{env}\dot{e}_z$, which can be estimated as $\hat{f}_{HC}\dot{e}_z$. With (10), (28) and (39), if $|e_z|>\epsilon$, the system energy flow is then:

$$\begin{aligned}\dot{V}+\dot{V}_{rob}&=\dot{V}+\hat{f}_{HC}\dot{e}_z\\&=-d_i(t)\dot{e}_z^2+f_r\dot{e}_z+\frac{1}{2}\dot{\bar{k}}_i(t)e_z^2\\&\le -d_i(t)\dot{e}_z^2+f_r\dot{e}_z+[d_i(t)\dot{e}_z^2-f_r\dot{e}_z+p_1]\\&=p_1.\end{aligned} \tag{40}$$

This proves that p_1 from (29) is essentially an upper bound of the human–robot system energy flow $\dot{V}+\dot{V}_{rob}$ when $|e_z|>\epsilon$, and this upper bound can be flexibly adjusted by λ.

Furthermore, consider the total energy W that incorporates the system energy (40) and the energy tank V_{tk} in (30) :

$$W=V+V_{rob}+V_{tk}. \tag{41}$$

The total energy flow $\dot{W}$ can be computed with (30) and (40):

$$\begin{aligned}\dot{W}&=\dot{V}+\dot{V}_{rob}+\dot{V}_{tk}\\&=-d_i(t)\dot{e}_z^2+f_r\dot{e}_z+\frac{1}{2}\dot{\bar{k}}_i(t)e_z^2\\&\quad+(d_i(t)-p_2)\dot{e}_z^2-f_r\dot{e}_z-\frac{1}{2}\dot{\bar{k}}_i(t)e_z^2\\&=-p_2\dot{e}_z^2\le 0\end{aligned} \tag{42}$$

which proves the passivity of MPVIC, under $d_i(t)$ from (26), and $\bar{k}_i(t)$ from (28). The total energy flow (42) also shows that the energy tank V_{tk} fully absorbs or provides the change of the human–robot system energy $V+V_{rob}$, and the tank itself slowly dissipates energy at the rate of $p_2\dot{e}_z^2$.

Finally, the case when the energy budget in V_{tk} is fully consumed is considered, i.e., $V_{tk}=0$, and one has $p_1=0$ from (29), indicating that the human–robot system energy $V+V_{rob}$ is non-increasing according to

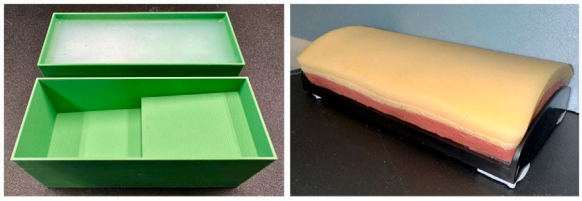


Fig. 5. Two silicone rubber models A (left) and B (right).

(42). Thus, any system variation by e_z or $\dot{e}_z$ will be compensated by the change of $\bar{k}_i(t)$ to maintain the non-increasing $V+V_{rob}$ until V_{tk} becomes positive again, or $\bar{k}_i(t)$ reaches zero leading to a safety stop due to s_{stop} in (31). □

4. Experiments and results

To evaluate the performance of the proposed control scheme, experiments have been conducted using a 7-DoF KUKA iiwa robot with a 2-finger Robotiq gripper where two XELA tactile sensors are attached to the gripper fingers, as shown in Fig. 1. In the experiments, the 3D-printed tools are held by the gripper and manipulated by the robot manipulator where the tactile sensors measure the contact force (i.e., the force along the normal direction of the environment). The control scheme is implemented by MATLAB and the tactile sensing information is communicated through ROS with a communication frequency of 100 Hz, and according to Remark 6, the linearization of MPC is essential to meet this communication frequency.

4.1. Experiments on silicone rubber models

To fully simulate the complex environment of the human tissues, two types of silicone rubber models as shown in Fig. 5 are utilized as the interaction environment in this work. The first silicone rubber model, namely Model A, is horizontally inhomogeneous with different depths, where the stiffness changes are gentle in general but with a subtle variation. The second model, namely Model B, is vertically inhomogeneous with multiple layers. These two models simulate the stiffness changes in different directions. The Young's modulus is measured as 14.50 kPa at $z = 5$ mm and 18.07 kPa at $z = 8$ mm for Model A, and 5.47 kPa at $z = 4$ mm and 7.91 kPa at $z = 8$ mm for Model B, roughly matching Young's modulus of human muscle (8 to 17 kPa) and skin (1 to 10 kPa) [40].

Two experiments on both models shown in Fig. 6 have been conducted. In Experiment 1, both models retract from the tool at a random time and remain in the retracted position until the task ends. In Experiment 2, the models are pushed against the tool. Moreover, for comparison purposes, the IC [4] and MPIC [25] are also tested. Table 2 lists the selected impedance model parameters for IC, MPIC, and MPVIC in the form of the following vector:

$$p_{imp} = [m_i, d_{i,min}, d_{i,max}, k_{i,max}, z_r, f_r] \tag{43}$$

The silicone rubber model type (Env.) is listed in Table 2 for Experiment 1 as well. The three approaches adopt the same impedance parameters in Experiment 2.

Considering the high inhomogeneity of interaction environments assumed in the experiments and the demand of real-time computation, the relatively short prediction horizon h_p and control horizon h_c for both MPIC and MPVIC are selected, which are $h_p = 5$ and $h_c = 3$ throughout the experiments.

It is experimentally supported that such low horizons have almost no negative effect on the interaction task performance thanks to the high control frequency (100 Hz) achieved by the linearization of MPC. The same acceleration constraints are set for the MPC in MPIC and MPVIC as $|\ddot{e}_z| < 5$. The boundary of the force change rate for MPVIC is selected as $\dot{e}_{fb} = 1$, and transition mode widths are $\omega_1 = 0.3$ and $\omega_2 = 0.5$ respectively. Passivity-related parameters are selected as $\lambda = 5$, $V_{tk,0} = 5$ and $p_2 = 0.001$. The disturbance bound for the environment estimator is decided as $\Delta = 0.03$. The threshold value ϵ is selected as $\epsilon = 10^{-4}$ in this work.

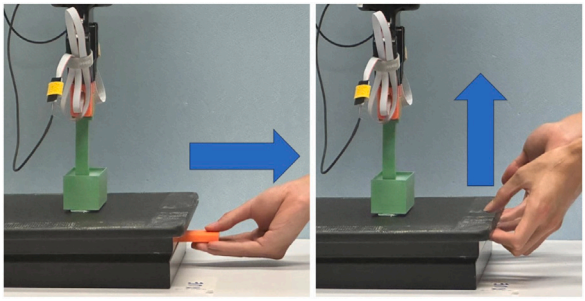


Fig. 6. Design of Experiment 1 (left) and 2 (right).

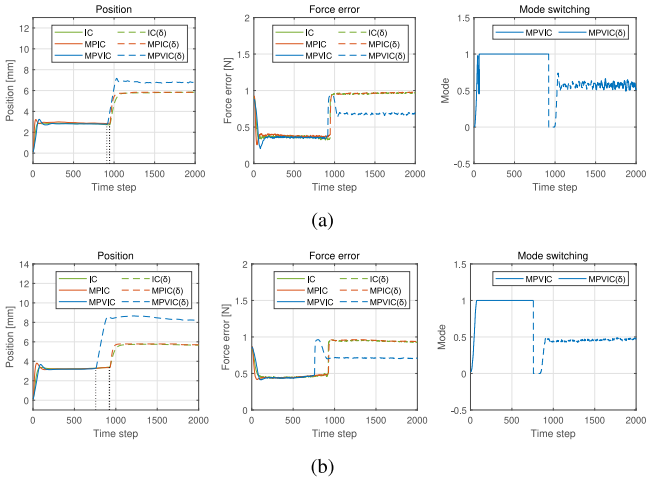


Fig. 7. Results of Experiment 1: (a) on Model A; (b) on Model B.

4.1.1. Experiment 1

Fig. 7 shows the results of Experiment 1 on both Models A and B. Here, dotted lines in the position plots indicate the periods while the disturbances are applied to the system (i.e., when retraction occurred), effectively illustrating the duration of these disturbances. Also, Table 2 lists the root-mean-square (RMS) and the maximum absolute (MaxA) of e_f [i.e., RMS errors (RMSEs) and MaxA errors (MaxAEs)] in Experiment 1 without and with disturbance, after the contact becomes steady.

As can be seen, the performances of all three control schemes are comparable in the absence of disturbances where the proposed control scheme slightly performs the best. This indicates that the real-time model estimation in the proposed MPVIC scheme provides an accurate model for MPC to adapt to the unknown/varying environment.

Significantly, the robots controlled by IC and MPIC lost contact with both silicone rubber models once they were retracted from the tool. These can be observed in Fig. 7 and Table 2, where $e_f \approx f_r$, indicating $f_{env} \approx 0$ (i.e., no contact force). In contrast, the proposed MPVIC-controlled robot can still maintain contact while it is subjected to environmental retraction/disturbance and thus it can achieve much better contact force control performance than the other two control schemes. In addition, the mode-switching mechanism of the proposed scheme works effectively on both models when the environments are retracted, with the mode switching from the performance mode to the transition or safety mode. Therefore, both performance and safety can be ensured by the proposed control scheme in Experiment 1.

Table 2
Experimental results of experiments 1.

Env.	p_{imp}	Disturbance	RMS of e_f [N]			Max of $\lvert e_f \rvert$ [N]		
			IC	MPIC	MPVIC	IC	MPIC	MPVIC
A	[0.01, 0.1, 0.4, 0.2, 1, 1]	No	0.366	0.375	**0.357**	0.393	0.409	**0.376**
B	[0.01, 0.1, 0.4, 0.2, 1, 1]	No	0.444	0.443	**0.439**	0.514	0.547	**0.506**
A	[0.01, 0.1, 0.4, 0.2, 1, 1]	Yes	0.958	0.894	**0.680**	0.974	0.902	**0.696**
B	[0.01, 0.1, 0.4, 0.2, 1, 1]	Yes	0.943	0.950	**0.711**	0.958	0.964	**0.720**

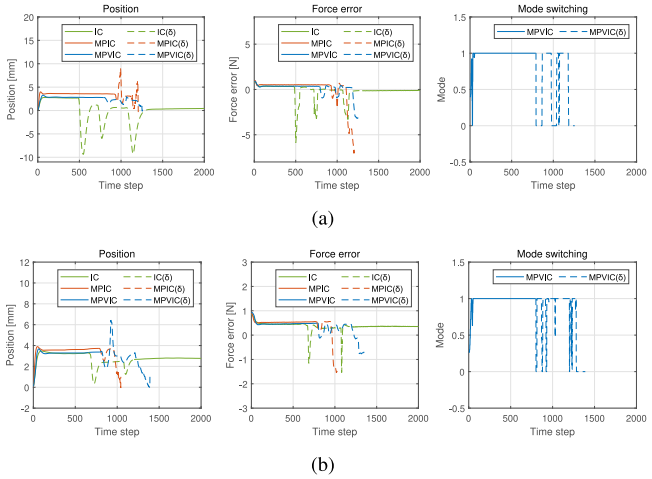


Fig. 8. Results of Experiment 2: (a) on Model A; (b) on Model B.

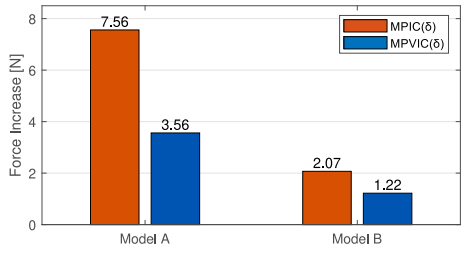


Fig. 9. Force increases in Experiment 2.

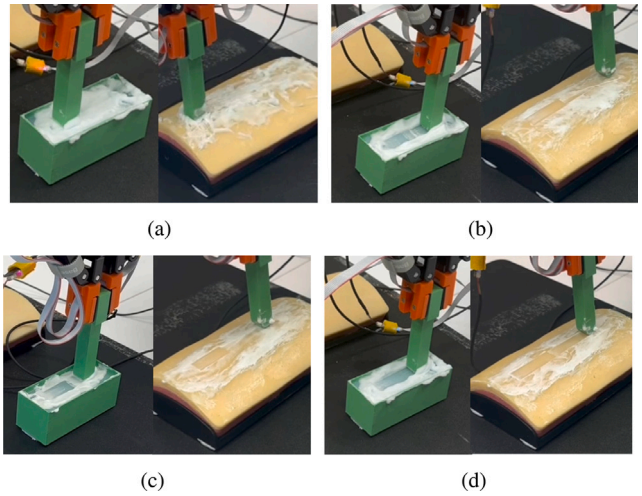


Fig. 10. Demonstration of three control schemes on the two models: (a) before start; (b) IC; (c) MPIC; (d) MPVIC.

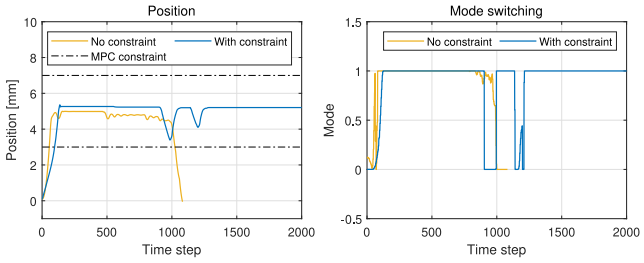


Fig. 11. Performance of MPVIC without and with constraints.

4.1.2. Experiment 2

Fig. 8 shows the results of Experiment 2 on both Models A and B. In this experiment, we focus on the reactions of MPIC and MPVIC upon disturbances by the interaction environment to demonstrate the effectiveness of the mode-switching mechanism. Similar to IC, both MPIC and MPVIC are compliant against small disturbances. For large and sudden disturbances, both MPIC and MPVIC stop the robot when the robot position exceeds the initial position in the case that the contact force model (9) is undefined when $z < 0$. Against such large and sudden disturbances, the robot controlled by MPIC complied with the environment change based on its fixed impedance model, while the robot controlled by MPVIC complied with the environment change via mode switching and its variable impedance model. This difference results in smaller average force increases for MPVIC compared to MPIC on both models as shown in Fig. 9. The average force increases for MPVIC are significantly small which are about 47% and 58% of the average force increases for MPIC on both models, respectively. This implies that better safety in terms of the contact force can be achieved by the proposed MPVIC.

In summary, both Experiments 1 and 2 demonstrate better performance and safety can be achieved by MPVIC upon IC and MPIC in the disturbance-rich and environment-changing interaction tasks. Significantly, the experimental results are consistent on both Models A and B for all three control schemes, indicating that the improved safety-performance balance of the MPVIC scheme is independent of model/environment properties. Fig. 10 visually demonstrates the performance of the three control schemes on both models in Experiment 1. The proposed MPVIC can consistently wipe the white gel more thoroughly than the other two control schemes. This is because the proposed MPVIC can successfully maintain the desired contact under unexpected disturbances that are larger than the bound Δ, while IC and MPIC fail.

Besides, the complexity of the three tested algorithms, IC, MPIC, and MPVIC, are compared via the average time consumption of 100 simulated interactions in MATLAB, each with 2000 time steps. The time consumptions of algorithms are recorded in Table 3. From the results, it can be concluded that the complexity of the proposed MPVIC is comparable with that of MPIC. Further improvements in computational complexity may include the simplified linear model for environment estimation, and the varying MPC horizons based on real-time performance.

4.2. Investigation on position constraints

Additionally, to emphasize the utility of constraints in the MPC, Experiment 3 is designed on Model A to compare the performance of MPVIC without and with position constraints. In this experiment, all

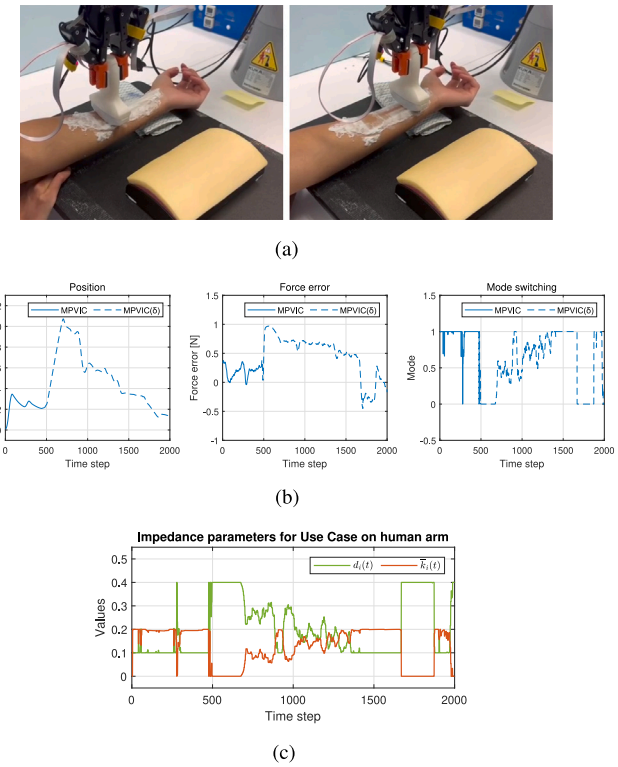


Fig. 12. Experiment on a human arm using MPVIC: (a) setup and demonstration; (b) results; (c) adaptive impedance parameters.

the parameters are the same as those in Experiment 2 on Model A, with similar disturbances (pushing against the tool). In this experiment, the constraints are set for position error as $|e_z| < 2$, while constraints can be readily applied to velocity as well (acceleration constraints unchanged). The position and force in the experiment are shown in Fig. 11. By comparing the performance of MPVIC without and with constraints, it can be observed that the robot is effectively constrained under disturbances, which ensures that the robot operates within a defined workspace and complies with the safety requirements. Therefore, the proposed MPVIC provides a safe and optimal solution for the interaction tasks where there are constraints to the robotic systems.

4.3. Case study on a human arm

Finally, being inspired by ultrasonography, an experiment has been conducted on a human arm using the proposed MPVIC, whose experimental setup is shown in Fig. 12(a). As can be seen, a 3D-printed ultrasound probe (as the tool) is grasped by the robotic gripper and manipulated by the robot manipulation. During the experiment, the probe is moved along the surface of the human arm, while the human arm retracts from and pushes against the probe with varying amounts at random times. These movements can be considered as unknown disturbances to the system, similar to the previous experiments. More specifically, the human arm retracts from the probe when the other hand under the arm moves away during the first phase, and the human arm moves up towards the probe in the later phase.

The experimental results of this interaction task on the human arm are shown in Fig. 12(b), and the variations of impedance parameters are also presented for better clarity in Fig. 12(c). It can be found that the robot controlled by the proposed MPVIC exhibits continued contact after the human arm retraction. Furthermore, the contact force remains within an acceptable range when the human arm pushes against the probe. The observation, based on the force error e_f, demonstrates that the MPVIC can establish stable contact with the human arm after the disturbances while ensuring safety. In conclusion, the proposed MPVIC can achieve safety-performance-balanced interaction in the unknown disturbance-rich environment.

Besides that, videos of some experiments presented in this Section can be found in the supplementary materials.

Table 3
Average time per simulation time step [milliseconds].

IC	MPIC	MPVIC
0.037	1.079	1.246

5. Conclusion

In this paper, a novel MPVIC was proposed to achieve safe and high-performance physical robotic interaction with unknown and disturbance-rich environments. The proposed MPVIC includes the real-time adaptive robust environment estimator to efficiently model the unknown environment with the nonlinear HC model. A mode-based VIC is designed to smoothly switch between safety and performance modes to achieve the target impedance model while ensuring safety upon abnormal interaction force or interaction force changing rate. An MPC-based framework incorporates the above sections to enable constraint compliance. The proposed MPVIC control scheme also has a comparable algorithm complexity to the state-of-the-art algorithm such as MPIC. It was also proven theoretically that the proposed parameter estimator could estimate the environment model against bounded disturbances, and the controlled system would remain passive against external energy injection from the physical interaction.

Several experiments have been conducted on two silicone rubber models and a human arm. The experimental results showed that the proposed control scheme outperformed the available control schemes in the interaction tasks with unknown environment contact models and bounded but unpredictable environment position shifts, such as robotic ultrasound scanning. Specifically, the proposed control scheme successfully maintained the contact compared to other available methods when the interaction environment suddenly retracted, and also enhanced interaction safety by a lower force increase when the environment drastically pushed against the robot. A case study on a real human arm with rich disturbance further confirmed the above advantages.

For future directions, learning-based tuning of hyperparameters in the control scheme and the incorporation of high-level task planning could be considered to reduce the requirement of manual tuning and enhance the readiness for real-world application. A generalized abnormality examiner and additional visual systems could also contribute to handling more generalized and complex real-world situations.

CRediT authorship contribution statement

Junyuan Xue: Writing – original draft, Validation, Methodology. **Wenyu Liang:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Yan Wu:** Writing – review & editing, Supervision, Conceptualization. **Tong Heng Lee:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.robot.2025.104961.

Data availability

Data will be made available on request.

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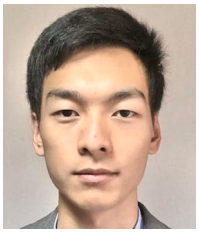
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